Annotation-Informed Models

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Annotation-informed Bayesian linear regression uses external marker annotations to structure priors over marker inclusion, effect variance, or mixture-component membership. The likelihood still comes from phenotype summary statistics and LD. Annotations change how prior probability and scale are distributed across markers before and during posterior sampling.

In `sblr`, the polished annotation workflows build on the CSR ST-BLR data contract:

```
stats + disk-backed sparse LD + aligned marker annotations
    -> stblr_csr_annot(annotation_model = ...)
    -> posterior marker effects, inclusion probabilities, and diagnostics
```

This note explains the implemented public wrappers and places them within the broader empirical-Bayes and multi-trait framework preserved in the archived annotation draft. Full marker-update derivations and implementation sketches belong in Technical annotation priors; full stick-breaking details belong in Technical SBayesRC-style annotations.

1 Why Annotations Inform Priors

Markers are not exchangeable with respect to existing biological knowledge. They may lie in coding regions, regulatory regions, QTL intervals, conserved sequence, tissue-specific chromatin states, or other externally defined sets. Annotation-informed models ask whether that information should alter a marker's prior probability of being active or the expected scale of an active effect.

For marker i and trait t, useful prior quantities include:

$$\pi_{it} = P(d_{it} = 1), \qquad v_{b,it} = \mathrm{Var}(b_{it} \mid d_{it} = 1),$$

and, for mixture models, component probabilities $\pi_{itc} = P(c_{it} = c)$. In proposed multi-trait extensions, annotations could additionally influence marker-specific sharing-model probabilities or effect covariance matrices.

Annotation information can enter at different stages:

- **fixed/external:** marker-specific priors are supplied before fitting;
- **empirical Bayes:** priors are estimated from an earlier genomic or annotation-level analysis, then treated as fixed;
- **direct learned:** annotation coefficients are updated inside the genomic sampler;
- **grouped:** markers share parameters within one mutually exclusive group;
- **mixture-based:** annotations determine probabilities of null and active effect-size components.

These approaches answer different questions and propagate different amounts of uncertainty. None makes annotation membership causal by itself.

2 Annotation Matrix, Overlap, Centering, and Marker Ordering

Let A be an $m \times K$ annotation matrix:

$$A = \begin{bmatrix} A_{11} & \cdots & A_{1K} \\ \vdots & & \vdots \\ A_{m1} & \cdots & A_{mK} \end{bmatrix},$$

where rows are markers and columns are annotations. Binary entries can encode set membership, while continuous columns can encode scores or measurements. Annotations may overlap: a marker can belong to several biological categories simultaneously.

Overlap is scientifically useful but complicates interpretation. A direct sum over each annotation double-counts markers in multiple sets and is not a strict variance partition. Conditional annotation models instead use the whole row A_i to estimate the contribution of one annotation while accounting for the others. Strongly correlated annotation columns can still make individual coefficients weakly identified.

All public annotation wrappers prepare `A` with the package's annotation matrix helper. It:

- requires one row per model marker;
- reorders named rows to marker names when all names are available;
- rejects non-finite values;
- can add an intercept column;
- can standardize non-binary columns;
- optionally centers and standardizes binary columns.

Row order remains a hard correctness condition:

```
row i of A
    = row i of stats$wy and stats$ww
    = row/column i of sparse LD
    = row i of fit$bm and fit$dm
```

The workflow makes this explicit:

```
marker_id <- Glist$rsidsLD[[chr]]
A <- sim$annot
rownames(A) <- marker_id
stopifnot(nrow(A) == length(marker_id),
          identical(rownames(A), marker_id))
```

Preparation of `stats`, sparse LD, and marker order is described in Sparse-LD and BED workflows.

Centering has a statistical role beyond numerical convenience. If $z_{it} = A_i \eta_t$, a centered predictor is

$$z_{it}^{\text{centered}} = A_i \eta_t - \frac{1}{m} \sum_{j=1}^{m} A_j \eta_t.$$

Combining a centered annotation predictor with a global prior tends to redistribute prior mass across markers rather than allowing annotations to change the overall sparsity or variance scale without constraint.

3 Plain/Global Prior Baseline

The baseline BayesC-style sparse prior gives markers a common active probability and active-effect variance:

$$d_{it} \sim \text{Bernoulli}(\pi_t), \qquad b_{it} \mid d_{it} = 1 \sim N(0, v_{b,t}), \qquad b_{it} \mid d_{it} = 0 = 0.$$

Equivalently,

$$b_{it} \sim (1 - \pi_t)\delta_0 + \pi_t N(0, v_{b,t}).$$

This is the comparison model for annotation-aware priors. If annotations have no effect, fixed marker priors reduce to common values, learned annotation coefficients remain near zero, all group parameters become similar, and SBayesRC-style component probabilities do not vary meaningfully by annotation profile.

Annotation models should therefore be compared with a plain CSR fit using compatible markers, LD, priors, and MCMC settings. Differences should not be attributed to annotations when the underlying data preparation or global architecture changed at the same time.

4 Fixed Marker-Specific Priors

`stblr_csr_prior_annot()` fits CSR ST-BLR using fixed marker-specific inclusion probabilities, fixed variance multipliers, or both. It supports two routes:

1. supply `fixed_pi_marker` and `fixed_vb_multiplier` directly; or
2. supply `A` with fixed coefficients `beta_pi` and `beta_vb`, from which the wrapper constructs marker-specific priors.

For fixed annotation coefficients, the implemented construction centers the annotation linear predictors:

$$\ell_{it}^{\pi} = A_i \beta_{\pi,t} - \frac{1}{m} \sum_j A_j \beta_{\pi,t},$$

then computes

$$\pi_{it} = \text{logit}^{-1} \left\{ \text{logit}(\pi_t) + \ell_{it}^{\pi} \right\},$$

bounded to `pi_min` and `pi_max`. The variance multiplier is

$$\lambda_{it} = \exp \left(A_i \beta_{v,t} - \frac{1}{m} \sum_j A_j \beta_{v,t} \right),$$

bounded to `vb_multiplier_min` and `vb_multiplier_max`, with $v_{b,it} = v_{b,t}\lambda_{it}$.

The direct route is illustrated by the annotation workflow:

```
fit_fixed <- stblr_csr_annot(
  stats = stats,
  Glist = Glist,
  annotations = list(
    A = A,
    fixed_pi_marker = sim$pi_marker,
    fixed_vb_multiplier = sim$vb_multiplier,
    use_pi_marker = TRUE,
    use_vb_multiplier = TRUE
  ),
  annotation_model = "prior"
)
```

The compatibility wrapper `stblr_csr_prior_annot()` accepts the same underlying prior information for explicit fixed-prior workflows.

Fixed priors are appropriate when prior values come from external biological knowledge, an independent study, cross-fitted estimates, or a deliberately separate empirical-Bayes stage. They are also the simplest way to test whether the sampler responds sensibly to known marker-specific prior structure.

The key limitation is uncertainty: once supplied, these values are treated as fixed during the genomic fit. If they were estimated from the same data, the final posterior does not automatically propagate that first-stage uncertainty.

5 Direct Learned Annotation Effects

`stblr_csr_learn_annot()` learns annotation effects while fitting marker effects. For each trait, it maintains annotation coefficient vectors $\eta_{\pi,t}$ and $\eta_{v,t}$. The implemented marker-specific prior is conceptually

$$\operatorname{logit}(\pi_{it}) = \operatorname{logit}(\pi_t) + \operatorname{center}_i(A_i\eta_{\pi,t}),$$

and

$$v_{b,it} = v_{b,t}\exp\{\operatorname{center}_i(A_i\eta_{v,t})\}.$$

In the second expression, the exponential term is a multiplier of the global effect variance. The implementation caps probabilities and multipliers:

$$\pi_{it} \in [\pi_{\min}, \pi_{\max}], \qquad \frac{v_{b,it}}{v_{b,t}} \in [\lambda_{\min}, \lambda_{\max}].$$

The sampler alternates between marker updates under current marker-specific priors and annotation-coefficient updates. At intervals controlled by `annot_update_every`, random-walk proposals perturb $\eta_{\pi,t}$ or $\eta_{v,t}$. The inclusion-coefficient target uses current inclusion states, while the variance-coefficient target uses current active effects. Gaussian shrinkage scales `sigma_eta_pi` and `sigma_eta_vb` regularize the coefficients; proposal scales `rw_sd_eta_pi` and `rw_sd_eta_vb` control Metropolis movement.

```r
fit_learned <- stblr_csr_annot(
  stats = stats,
  Glist = Glist,
  annotations = A,
  annotation_model = "learned",
  learn_pi_annot = TRUE,
  learn_vb_annot = TRUE,
  rw_sd_eta_pi = 0.02,
  rw_sd_eta_vb = 0.02,
  annot_update_every = 10
)
```

The fit reports posterior annotation coefficients as `eta_pi` and `eta_vb`. They should be interpreted jointly with annotation scaling, coefficient shrinkage, bounds, proposal behavior, marker-level posterior summaries, and convergence diagnostics.

The compatibility wrapper `stblr_csr_learn_annot()` remains available when a workflow needs the lower-level learned-annotation call explicitly.

Use the learned model when annotation effects are genuinely part of the inferential target and there is enough information to estimate them. It is more coherent than treating same-data estimates as fixed, but also more demanding: overlapping annotations, weak signals, poor proposal scales, and short chains can produce unstable or poorly mixed coefficient estimates.

6 Group Priors

`stblr_csr_group_annot()` assigns every marker to exactly one group and gives each group trait-specific prior parameters. If $g(i)$ is marker i's group index, the prior is

$$d_{it} \sim \text{Bernoulli}(\pi_{g(i),t}), \qquad b_{it} \mid d_{it} = 1 \sim N\left(0, v_{b,t}\lambda_{g(i),t}\right).$$

The R wrapper accepts a length-m `group` vector, optionally named by marker. It converts group labels into an internal zero-based `group_index`, records group names and sizes, initializes `group_pi` and `group_vb_multiplier`, and returns group-level posterior summaries including included-marker counts.

```r
group <- ifelse(A[, "Anno1"] != 0, "annotated", "background")
names(group) <- rownames(A)

fit_group <- stblr_csr_annot(
  stats = stats,
  Glist = Glist,
  annotations = group,
  annotation_model = "group",
  group_pi_init = c(0.002, 0.001),
  group_vb_multiplier_init = c(1.1, 1.0),
  updatePi = TRUE,
  updateGroupVb = TRUE
)
```

The compatibility wrapper `stblr_csr_group_annot()` exposes the same grouped backend for workflows that prefer an explicit group-prior function.

Group inclusion probabilities have group-specific beta-prior controls. Variance multipliers can be updated and optionally normalized. Small groups need especially strong regularization because their inclusion counts and variance estimates are noisy.

Group priors are useful when categories are naturally mutually exclusive or when a simple, auditable model is preferred. They are also a practical simplification of overlapping annotations: define a deterministic hierarchy or a small set of combined categories, then fit one group per marker. That simplification discards overlap structure and makes results depend on the assignment rule. When overlap itself is scientifically important, learned matrix-based or SBayesRC-style models are usually more appropriate.

7 SBayesRC-Style Overlapping-Annotation Mixture Priors

`stblr_csr_sbayesrc_generic()` replaces a single active-effect distribution with multiple variance components. Let

$$\gamma = (\gamma_1, \dots, \gamma_C), \qquad \gamma_1 = 0, \quad \gamma_c > 0 \text{ for } c > 1,$$

where the zero component is null and active components scale the marker-effect variance. A common workflow choice is `gamma = c(0, 0.01, 0.1, 1)`.

This is an **SBayesRC-style annotation-dependent BayesR prior**. Its core prior family is closely aligned with the published SBayesRC prior structure: marker effects follow a BayesR-like mixture, and marker-specific component probabilities use conditional probit stick-breaking regressions on annotations. It is not an exact reproduction of GCTB SBayesRC as exposed by default. The public wrapper uses the four-component gamma grid above, whereas published SBayesRC uses `c(0, 0.001, 0.01, 0.1, 1)`. In addition, `sblr` uses a CSR sparse-LD likelihood rather than GCTB's low-rank rotated summary-statistics likelihood. Users can explicitly request the published gamma grid:

```
fit_sbayesrc_published_gamma <- stblr_csr_annot(
  stats = stats,
  Glist = Glist,
  annotations = A,
  annotation_model = "sbayesrc",
  gamma = c(0, 0.001, 0.01, 0.1, 1)
)
```

Matching the gamma grid aligns that part of the prior specification; it does not make the complete implementation identical to GCTB SBayesRC.

For marker i, annotations determine component probabilities through a generalized probit stick-breaking construction. Let

$$\eta_i = A_i \alpha, \qquad p_{ij} = \Phi(\eta_{ij}),$$

for $C - 1$ stick-breaking steps. Then

$$\begin{aligned}
\pi_{i1} &= 1 - p_{i1}, \\
\pi_{i2} &= p_{i1}(1 - p_{i2}), \\
&\vdots \\
\pi_{iC} &= \prod_{j=1}^{C-1} p_{ij}.
\end{aligned}$$

The probabilities are non-negative and sum to one. The native sampler floors and renormalizes them for numerical stability. Marker i's expected gamma multiplier is

$$E(\gamma_i \mid A_i, \alpha) = \sum_{c=1}^{C} \pi_{ic} \gamma_c.$$

`make_sbayesrc_alpha_init()` creates deterministic initial values. It distributes the initial active probability across non-null components, converts those marginal probabilities into conditional stick-breaking probabilities, and, when an all-ones intercept exists, encodes the baseline in that intercept row of `alpha_init`. Other coefficients begin at zero unless supplied.

```
fit_sbayesrc <- stblr_csr_annot(
  stats = stats,
  Glist = Glist,
  annotations = A,
  annotation_model = "sbayesrc",
  gamma = c(0, 0.01, 0.1, 1),
  updateAlpha = TRUE,
  alpha_update_every = 10
)
```

The fit contains annotation coefficients `alpha`, annotation-effect variances `sigmaSqAlpha`, marker component probabilities `comp_prob`, and component counts `ncomp`.

CSR SBayesRC also supports BayesS-style MAF-dependent prior variance scaling through `selection_s`. Fixed global `selection_s` and sampled trait-specific `selection_s` are supported for `annotation_model = "sbayesrc"`:

```
fit_sbayesrc_sampled_s <- stblr_csr_annot(
  stats = stats,
  Glist = Glist,
  annotations = A,
  annotation_model = "sbayesrc",
  estimate_selection_s = TRUE
)
```

In this model, annotations affect component probabilities and `selection_s` affects marker-specific effect-size prior variance. Sampled `selection_s` is not supported by the BayesC-like annotation-aware CSR backends `annotation_model = "prior"`, `"learned"`, or `"group"`. See selection_s implementation status for the scale convention, sampled-S defaults, and backend support matrix.

The exported diagnostic helpers convert fitted coefficients into compact, deterministic summaries:

- `sbayesrc_annotation_pi(alpha, gamma)` applies stick-breaking to each annotation coefficient row;
- `sbayesrc_annotation_gamma_mean(alpha, gamma)` returns expected gamma per annotation row;
- `sbayesrc_marker_pi(A, alpha, gamma)` applies stick-breaking to $A\alpha$, returning marker-level component probabilities;
- `sbayesrc_marker_gamma_mean(A, alpha, gamma)` returns expected gamma per marker.

Annotation-row diagnostics isolate one coefficient row and are not the same as marker-level predictions under overlapping annotation profiles. For biological interpretation, inspect both, and compare marker expected gamma with posterior inclusion probabilities without assuming they should be identical.

Use the SBayesRC-style model when annotations may overlap and the scientific question concerns how annotations shift probability among null, small, moderate, and larger effect components. It is more expressive than a binary active/null model and correspondingly needs careful prior, mixing, and identifiability diagnostics.

8 Annotation Workflow

The current workflow connects simulation, CSR preparation, four models, and compact diagnostics:

```
mtsim_annotation()
    -> summarize_annotation_signal()
    -> make_stats() + make_sparseLD()
    -> align A with stats and sparse LD
    -> stblr_csr_annot(annotation_model = "prior")
    -> stblr_csr_annot(annotation_model = "learned")
    -> stblr_csr_annot(annotation_model = "group")
    -> stblr_csr_annot(annotation_model = "sbayesrc")
    -> compact fit, annotation, and expected-gamma summaries
```

`mtsim_annotation()` simulates phenotypes, marker effects, overlapping annotations, enriched causal-marker sampling, and marker-specific prior values. It supports known-truth checks but does not represent a real-data validation by itself.

`summarize_annotation_signal(sim)` reports annotation sizes, causal counts, and causal rates. With a fit, it can add trait-specific mean posterior inclusion and mean absolute posterior effect summaries. These summaries are descriptive: annotation overlap and LD mean they are not additive variance partitions.

The workflow then constructs `stats` and updates `Glist$sparseLD` from the same BED marker order, verifies `A`, and fits all four prior families through `stblr_csr_annot()`. Full marker matrices remain available in fit objects, while compact summaries prevent marker-scale output from flooding the console. See Workflows and outputs for reporting patterns.

9 Empirical-Bayes Annotation Workflows

The archived annotation draft describes a broader staged strategy:

genomic STBLR/MTBLR $\to$ marker-level posterior summaries $\to$ annotation-level model $\to$ annotation-informed

First-stage summaries can include posterior means $\hat{b}_{it}$, posterior inclusion probabilities $\widehat{dm}_{it}$, residual scores, and set-specific genomic variance contributions. For an annotation set S_k, a smoothed inclusion estimate is

$$\hat{\pi}_{k,t} = \frac{\alpha + \sum_{i \in S_k} \widehat{dm}_{it}}{2\alpha + |S_k|}.$$

An effect-scale estimate can use the estimated set contribution

$$\widehat{V}_{G,k,t} = \frac{1}{n} \sum_{i \in S_k} \hat{b}_{it}(wy_{it} - r_{it})$$

and divide by an effective number of active markers,

$$m_{k,t}^{\text{eff}} = \sum_{i \in S_k} \widehat{dm}_{it}, \qquad \hat{\sigma}_{b,k,t}^2 = \frac{\widehat{V}_{G,k,t}}{\max(m_{k,t}^{\text{eff}}, c)}.$$

For overlapping annotations, marker-level responses such as $\widehat{dm}_{it}$, $\hat{b}_{it}^2$, or estimated variance contributions can be regressed jointly on A. The resulting coefficients can construct marker-specific priors on logit or log scales.

This strategy preserves useful technical ideas for prior learning, annotation-level STBLR/MTBLR, multi-trait sharing priors, and warm starts. However, it is not identical to the four implemented direct workflow wrappers. When the same data estimate priors and evaluate the final model, uncertainty can be understated. Sample splitting, chromosome-wise cross-fitting, external estimates, or a fully hierarchical model are more conservative alternatives.

10 Choosing an Annotation Model

Model	Use when	Main advantage	Main caution
Fixed marker-specific prior	Reliable external, cross-fitted, or deliberately staged priors already exist; or a known-prior simulation is being checked.	Stable and auditable; simplest marker-specific model.	Treats prior values as known and does not propagate their estimation uncertainty.
Learned annotation coefficients	Inclusion or effect-scale annotation effects are inferential targets and sufficient signal exists to estimate them.	Learns annotation effects jointly with marker effects.	Requires careful shrinkage, proposal tuning, bounds, and convergence assessment.
Group prior	Markers belong to one meaningful category each, or a simple mutually exclusive approximation is acceptable.	Easy to interpret and computationally direct.	Discards or arbitrarily resolves overlapping annotations; small groups are unstable.
SBayesRC-style mixture prior	Overlapping annotations should shift probability among null and multiple active-effect scales.	Flexible effect-size architecture with marker-level mixture probabilities.	More parameters and more difficult identifiability and mixing diagnostics.

Model	Use when	Main advantage	Main caution
Empirical-Bayes staged prior	First-stage genomic evidence should inform later fits or proposed MTBLR extensions.	Flexible bridge between posterior summaries and future priors.	Same-data reuse can be optimistic; many draft components remain research-stage.

Start with the simplest model that matches the scientific question. More expressive annotation models do not automatically produce more reliable biological conclusions.

11 Regularization, Caps, Shrinkage, and Overfitting

Annotation-informed priors can become aggressive because probabilities and variance scales enter marker updates repeatedly. The following controls are part of the model, not merely numerical safeguards:

- center annotation linear predictors around their marker average;
- standardize continuous annotations so coefficient scales are comparable;
- decide deliberately whether binary annotations should remain 0/1 or be centered and standardized;
- shrink annotation coefficients toward zero;
- bound marker inclusion probabilities away from zero and one;
- cap variance multipliers to avoid extreme marker scales;
- use pseudo-counts and shrinkage for small annotation sets;
- inspect correlated annotation columns and unstable coefficient directions;
- use cross-fitting or external estimates for formal empirical-Bayes inference.

The default bounds in current wrappers are intentionally broad and are not scientific defaults for every analysis. Sensitivity to bounds, prior scales, annotation preprocessing, and LD construction should be reported.

12 Interpretation and Causal Limits

Annotation-informed models estimate how annotation structure relates to prior architecture and posterior genomic evidence under the fitted model. They can identify annotations associated with higher posterior inclusion, different effect scales, or different mixture-component probabilities. They can support biological prioritization and generate hypotheses.

They do **not** prove that an annotation, gene, tissue, or regulatory mechanism causes a phenotype. Apparent enrichment can reflect:

- LD between annotated and causal markers;
- marker density and annotation size;
- correlated or overlapping annotations;
- phenotype architecture and model misspecification;

- sparse-LD approximation and marker filtering;
- winner's curse or reuse of the same data to estimate and evaluate priors.

Genetic sharing across traits is also not causal direction. Colocalization, Mendelian randomization, functional experiments, and external replication ask different questions and require their own assumptions.

Row alignment is essential: a misordered annotation matrix produces a well-formed but biologically meaningless prior. Short demonstration chains are only execution examples. Real analyses need longer chains, convergence and proposal diagnostics, sensitivity analysis, posterior predictive checks, and clear documentation of annotation preprocessing and prior construction.

13 Related Notes and Sources

- Model overview
- Sparse-LD and BED workflows
- Workflows and outputs
- Multi-trait models
- Technical annotation priors
- Technical SBayesRC-style annotations
- Complete annotation workflow
- Archived annotation-prior draft